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Sensorless load angle control for two-phase hybrid stepper motors☆

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ABSTRACT

The widely used full-step open-loop stepping motor drive algorithms are characterised by a low torque/power ratio, large torque ripple and resonance problems. In order to use stepping motors in more demanding applications, closed-loop control is needed. In this paper, a previously described load angle estimation algorithm, solely based on voltage and current measurements, is used to provide the necessary feedback without using a mechanical position sensor. This paper introduces a novel closed-loop load angle controller which optimises the current level based on the estimated load angle. This approach is challenging as the current - load angle dynamics are highly nonlinear. However, in this research a linear model is proposed and used to tune a PI load angle controller. The controller reduces the current level to obtain the optimal load angle but does not change the way stepping-motor are driven by step command pulses. This results in a broad industrial employability of these algorithms. The proposed sensorless controller is validated trough both simulations and measurements.

1. Introduction

The two-phase hybrid stepper motor principle is illustrated in Fig. 1 [1,2]. The stator is equipped with concentrated windings while the multitoothed rotor is magnetised by means of permanent-magnets. The rotor teeth are attracted by the excited stator phase. When a new full-step command pulse is given, the excitation of one phase is released while the second phase is excited. In half- and micro-stepping algorithms (Fig. 2), two-phases are excited simultaneously in order to increase the number of discrete rotor-position steps in a single revolution.

By counting the step command pulses, open-loop positioning is achieved. The fact that an expensive position sensor is not needed, makes these motors very appealing for industrial and domestic applications. However, the widely used open-loop drive algorithms are characterised by a low torque/power ratio, large torque ripple and resonance problems [3,4]. The basic open-loop algorithms are unsatisfactory to drive a stepper motor, used in more demanding applications. For this purpose, vector-control algorithms, as used in permanent-magnet synchronous machines (PMSM) and induction machines (IM), are of interest. Literature provides numerous research results concerning these algorithms for PMSM [5,6] and IM [7,8]. Some attempts are made to use these principles to drive stepper motors [9]. Literature also mentions research efforts concerning stepper motor control algorithms in order to reduce the torque ripple [10]. This is possible by adapting both the current level and waveform [11] or by optimising the commutation [12]. These stepping motor algorithms result in an enhanced performance of the stepping motor systems.

Determining the current commutation by means of step command pulses is very common in stepper motor applications. However, the advanced stepping motor drive algorithms described in [9,11,12] use control loops typical for PMSM machines. This means positioning by using step command pulses is impossible when the methods described in [9,11,12] are implemented. This hinders the implementation of these methods in industry. The controller proposed in this paper only adapts the current level. In this way, the described method is complimentary to the popular method used to drive a stepper motor, which is based on step command pulses. This contributes to the industrial employability of the algorithm.

The methods described in [9,11] need position feedback to work properly. Using a mechanical position sensor to obtain the necessary feedback for the closed-loop control is not a preferable option because it increases the cost and complexity of the system. Therefore, a rotor-position estimator is a more favorable option to provide feedback for vector-control operation. Such estimators are often referred to as sensorless and self-sensing methods. A generic view on the current state-of-the-art in estimation based on measurement data given in [13,14]. Particularly, recent advances in research concerning sensorless control for PMSM [15,16] and IM

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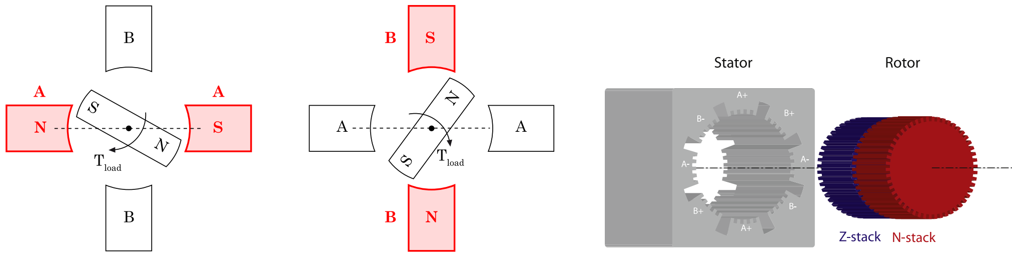


(a) Full-Step principle (b) Hybrid stepper motor construction

Fig. 1. Stepper motor principle.

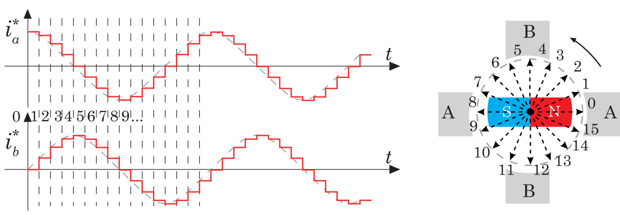


Fig. 2. Micro-stepping principle.

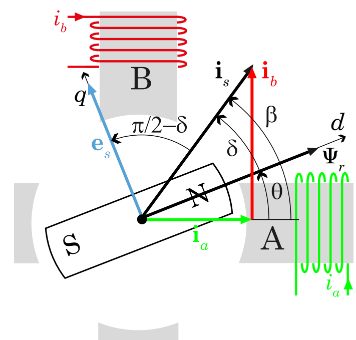


Fig. 3. Back-EMF $\mathbf{e}_s$, current $\mathbf{i}_s$ and flux $\boldsymbol{\Psi}_r$ vectors and load angle δ.

[17,18] are of interest. Research efforts has also been made to implement sensorless control for stepper motors [19–21].

The majority of these sensorless methods [17,18,20,21] is based on observer techniques and Kalman filtering. The tuning of such observers is complex and time consuming. Moreover, most observer-based sensorless techniques [17,20,21] rely on models containing load parameters such as inertia and friction. These load dependent parameters are difficult to determine and often vary over time. On the other hand, [15,16,21] estimate the rotor position based on the response of high frequency test signals. These methods estimate the rotor position accurately, even at low speeds where observer-based methods often fail. However, these methods need acces to the power electronics to generate the desired test pulses. This hinders implementation at existing stepping motor controllers.

The sensorless algorithm, described in [22,23], estimates the load angle. In contrast to more complex observer algorithms this estimator does not depend on mechanical load parameters and is characterised by a low computational cost. Because of these practical advantages, this algorithm is used in this paper to provide the feedback. Measurements are included in this paper to prove the accuracy of the estimator. However, the main added value of this paper is in the design of a control which adapts the stepping motor current level based on the estimated load angle. An important advantage of this approach is the fact that optimal performance can be obtained without changing the control architecture for the stepping motor user. The latter means the user can still control the position by sending and counting step command pulses while the control introduced in this paper continuously optimises the current level.

The load angle and current stepper motor driving principles are briefly discussed in Section 2. The load angle estimator introduced in [22,23] is briefly discussed in Section 3. This section also includes measurements to validate the estimator approach. Section 4 discusses the model describing the dynamic relationship between current level and load angle. While there is many literature on dynamic stepping motor models [1,2,10,20] the impact of the current level on the load angle is not described yet. This dynamic behaviour is highly nonlinear but a linearization is proposed in this paper. The model is used to obtain insight in the parameters influencing the load angle control. The linearized model is used to tune the PI load-angle controller as discussed in Section 5. Simulations, discussed in Section 6, are used to analyse the robustness of the controller design against user settings and mechanical load variations. Finally in Section 7 the steps needed for practical implementation are presented together with measurements validating the proposed sensorless controller.

2. Load angle

The electromagnetic torque generated by a hybrid stepper motor can be written as [2,24]:

$$T_{\text{motor}} = \Psi_r.i_s.\sin(\delta) + \frac{L_d - L_q}{2}.i_s^2.\sin(2\delta) \tag{1}$$

The load angle δ between the stator current vector $\mathbf{i}_s$ and the rotor flux $\boldsymbol{\Psi}_r$, illustrated in Fig. 3, determines the torque current ratio as indicated in (1).

The first term in (1) describes the torque generated by the interaction between the permanent-magnet rotor flux $\boldsymbol{\Psi}_r$ and the stator current $\mathbf{i}_s$. This term depends on the sine of the load angle δ. Because of the multitoothed rotor and stator construction of a hybrid stepper motor, the inductance along the direct axis L_d and the one along the quadrature axis L_q differs. This means that the reluctance effect increases the maximum electromagnetic torque. This reluctance effect is represented by the second term in (1) and varies sinusoidally by twice the load angle δ. To quantify both effects, a test motor with specifications given in Table 1, is used.

Table 1
Stepper motor specifications.

Dimension	NEMA23
# Rotor teeth p	50
Rated current $i_{\max}$	4.7A
Holding torque $T_{\max}$	0.75 Nm
Maximum speed	700 rpm
Phase resistance R_s	0.5 Ω
Phase inductance L_s	1 mH
Rotor inertia J	$2.2.10^{-5}$kgm^2

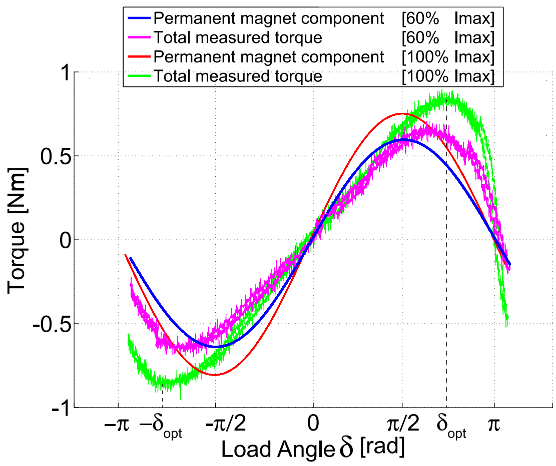


Fig. 4. Measured torque - load angle relation for 60% and 100% nominal current $I_{\max}$.

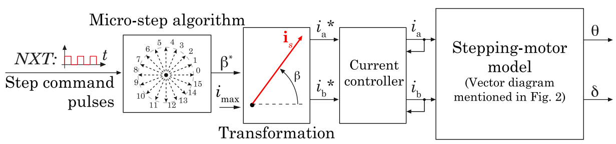


Fig. 5. Stepper motor drive principle where current vector $\mathbf{i}_s$ postion β is determined by step command NXT pulses sent by the user.

While the rotor is blocked the load angle δ is modified by changing the phase current setpoints i_a^* and i_b^*. For a current amplitude of 60% and 100% of the nominal current, torque measurement results are given in Fig. 4[1].

Both (1) and Fig. 4 illustrate the impact of the load angle on the torque generation. The required motor torque can be realised by both a high current–low load angle or low current–high load angle combination. The latter is more efficient. However, when the load angle exceeds $+\delta_{\text{opt}}$ the generated motor torque drops, resulting in a loss of synchronism between the step command pulses, sent by the user, and the actual rotor position.

For a given rotor position θ the load angle can be controlled by adapting the stator current vector $\mathbf{i}_s$ as illustrated in Fig. 3. This vector can be manipulated both in amplitude i_s and angular position β. However, the bulk of the stepper motor systems are driven in open-loop where the angular position β of the stator current vector $\mathbf{i}_s$ is determined by step command pulses as depicted in Fig. 5. To leave this method, used in industry, unadapted the current level i_s or magnitude of the stator current vector is used by the controller to obtain the load angle setpoint.

3. Load angle estimation

Using Lenz's law the back-EMF voltage vector $\mathbf{e}_s$, induced by the rotor flux $\mathbf{\Psi}_r$ for a certain number of windings N, can be written as:

$$\mathbf{e}_s = N.\frac{\mathrm{d}\mathbf{\Psi}_r}{\mathrm{d}t} \tag{2}$$

This implies a phase lead of $\pi/2$ rad between the back-EMF vector $\mathbf{e}_s$ and the flux vector $\mathbf{\Psi}_r$. From Fig. 3 it follows that the angle between the current vector $\mathbf{i}_s$ and the back-EMF vector $\mathbf{e}_s$ is $\pi/2 - \delta$. Since the current is easily measured, estimating the load angle is reduced to a problem where the back-EMF signal has to be estimated.

The stator phases are modeled as [2]:

$$u_s = R_s.i_s + L_s.\frac{\mathrm{d}i_s}{\mathrm{d}t} + e_s \tag{3}$$

When the stator phases are modelled, the position dependency of the inductances [1] caused by the multitoothed rotor and stator construction (Fig. 1) could be included. However, [1] suggests to neglect the inductance variation when the electrical dynamics are considered. Moreover, literature concerning stepper motor control [10,21] uses a constant inductance L_s and neglects the difference between L_d and L_q when describing the electrical behaviour of the stator phases.

[23] suggests to write (3) in the frequency domain:

$$E_s(j\omega) = U_s(j\omega) - j\omega.L_s.I_s(j\omega) - R_s.I_s(j\omega) \tag{4}$$

Simulations and measurements discussed in [23] show that the load angle can be estimated based on the fundamental waveform of the electrical signals. Fig. 6a illustrates the fundamental current signal derived from the measured current. Eq. (4) can be written at fundamental pulsation ω_1. As mentioned before, the step command pulses determine the rotational speed by adapting the current setpoints (see Fig. 2) from which the fundamental pulsation ω_1 is easily determined.

In [23] it is suggested to use the sliding discrete fourier transform (SDFT) [25–27] to calculate the fundamental voltage U_{s1} and current I_{s1} components based on consecutive measurement samples (shown in Fig. 6a) in a computationally efficient manner. For a period of N samples the z-domain transfer function of the kth order SDFT is:

$$\frac{1 - z^{-N}}{1 - e^{j2\pi.\frac{k}{N}}.z^{-1}} \tag{5}$$

A possible implementation of (5) is illustrated in Fig. 6b. When (5) is used to determine the fundamental current and voltage components I_{s1} and U_{s1} based on consecutive current and voltage measurement samples, (4) can be used to determine the fundamental back-EMF E_{s1}. According to Fig. 3 the estimated load angle $\hat{\delta}$ can then be written as:

$$\hat{\delta} = \frac{\pi}{2} - (\angle(E_{s1}) - \angle(I_{s1})) \tag{6}$$

As illustrated in (5) and Fig. 6b regardless of the window length N, the SDFT only needs one real sum, one complex sum and one complex multiplication to compute a DFT output. Altogether, estimating the load angle based on one stator phase can be done calculating only four real sums, three complex sums, four complex multiplications and determining the angle of a complex value twice. Thanks to this limited number of mathematical operations, this estimator is a competitive alternative for more complex observer algorithms [20,21,28].

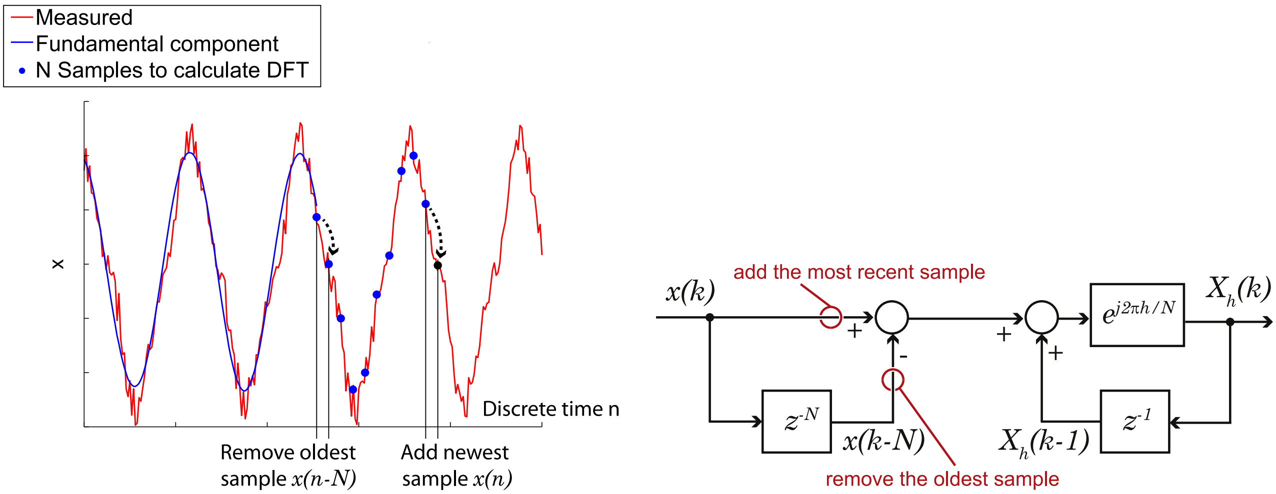
In contrast to many observer algorithms the proposed method does not need additional information concerning the mechanical load. Moreover, there is no tuning needed to obtain a correct estimator. The algorithm used in this paper only needs a correct value for the phase inductance and resistance.

3.1. Measurements

To validate the proposed estimation approach, the algorithm is implemented in dSpace at the test-rig depicted in Fig. 7.

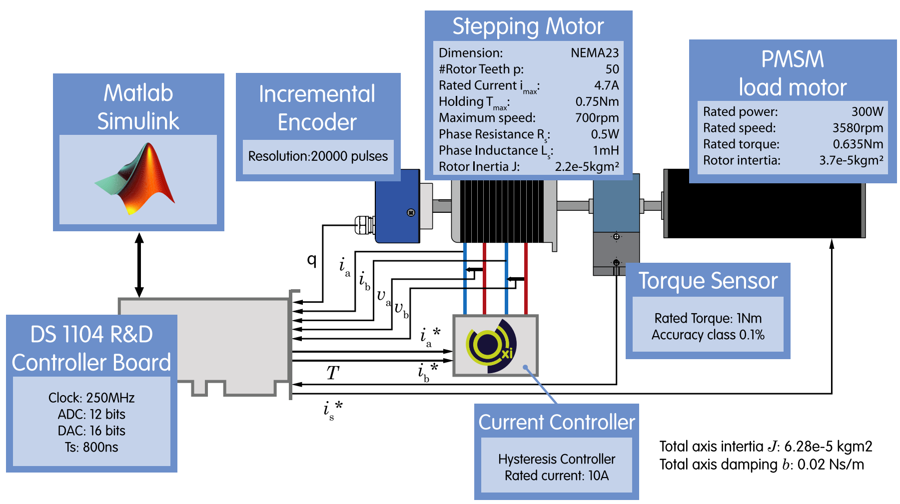
The measurement results depicted in Fig. 8a and b prove the capability of the SDFT algorithm to estimate the load angle. Furthermore, to illustrate the behaviour of the estimator during transients, a sudden change in load torque is applied during the measurements in both Fig. 8a and b. To drive the stepper motor at 100

[1] All angular values are given in electrical radians. For a machine consisting of p pole pairs this means the actual mechanical angle $\theta_{\text{mech}} = \theta/p$

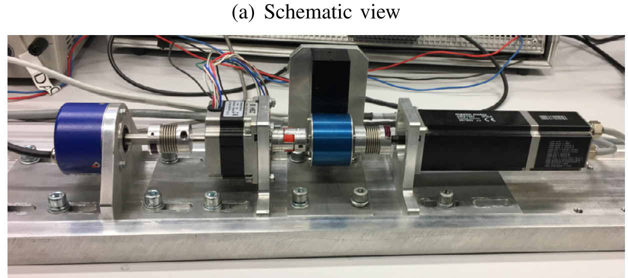


(a) (b)

Fig. 6. The sliding discrete fourier transform principle (a) and implementation (b) [25] .



(a) Schematic view



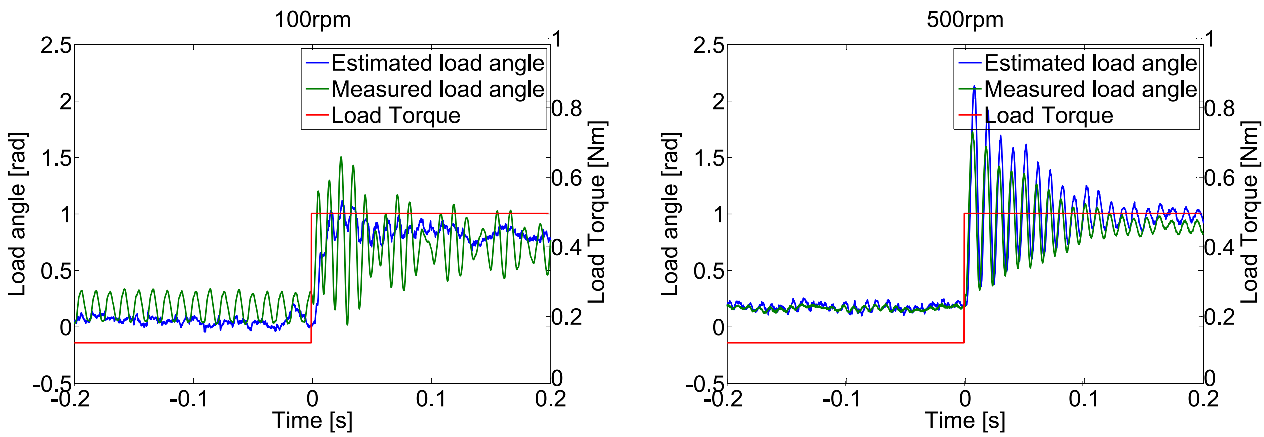
(b) Test rig photograph

Fig. 7. Test rig.

rpm, the drive current and voltage signals have a period of 12 ms. When the speed is increased to 500 rpm the signal period becomes 2.4 ms. When using a SDFT, the load angle estimation is updated with every new measurement sample. However, the estimation is based on all the samples within the sliding window, as illustrated in Fig. 6a. This sliding window has the same length as the signal period meaning the load angle estimation responds quicker at higher speeds. This dynamical behaviour is evaluated together with the process dynamics in Section 4 to acces the possible influence of the estimator on the control.

3.2. The estimation error

The speed dependent dynamics of the estimator explain why the algorithm gives detailed and correct estimations of the load angle oscillations at 500 rpm (Fig. 8b) but fails at 100 rpm (Fig. 8a). However, the estimator gives a correct view on the average load angle at both high and low speeds. Moreover, a correct estimation of the load angle oscillations is more crucial at higher speeds as these speed setpoints are closer to the operating and stability limits of the machine.



(a) Torque disturbance at $n = 100$ rpm [23]

(b) Torque disturbance at $n = 500$ rpm [23]

Fig. 8. Estimated and measured load angle at speed and torque disturbances.

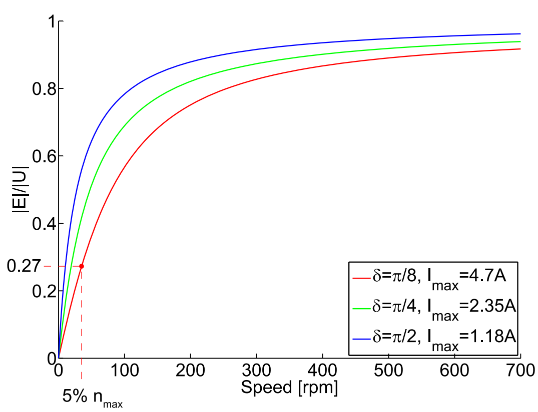


Fig. 9. Ratio between the back-EMF amplitude $|E_{s1}|$ and voltage vector amplitude $|U_{s1}|$ as a function of the rotational speed.

Secondly, the estimation algorithm is based on the back-EMF signal which is proportional to the rotor speed. At low speeds the amplitude of the back-EMF is small compared to other signals and inevitable measurement noise. To analyse the minimum speed, the stepping motor used in the measurement set-up depicted in Fig. 7 is considered. The ratio between the back-EMF amplitude $|\mathbf{E}_{s1}|$ and voltage vector amplitude $|\mathbf{U}_{s1}|$ is determined and shown in Fig. 9 for different load angle current combinations. As depicted in Fig. 9, for a situation with a low load angle and a high current, the back-EMF is smaller compared to the supply voltage amplitude. However, even at $\delta^* = \pi/8$ and a rotational speed of 5% of the maximum, the back-EMF amplitude is already 27% of the current vector amplitude. This means the algorithm delivers reliable estimations from speeds setpoints of only 5% of the maximum rotational speed.

An estimation error can be reduced by two possible means. First of all, Eq. (4) can be evaluated for both phase a and b. This means two independent load angle estimations are obtained for a two-phase stepping motor. A deviation between these two estimations indicates a possible estimation error. Furthermore, as the estimator depends on correct values for the stator resistance R_s and stator inductance L_s an online identification of these parameters is an enhancement of the proposed estimator. Numerous papers [29,30] describe the reliable identification of stator resistance and inductance. As the focus and main added value of this paper is in control, estimation errors should be avoided. Therefore the stator resistance R_s and stator inductance L_s used in this paper are identified by a least squares algorithm minimising the error between the measured load angle and the estimated load angle over the complete operating range. In this way, estimation errors do not distort the results in this paper.

4. Dynamic model

As mentioned in the last paragraph of Section 2, the current level i_s or magnitude of the stator current vector is used by the controller to obtain the load angle setpoint. Therefore this section discusses the dynamics of the load angle based on the current level i_s. There is many literature on dynamic stepping motor models [1,2,10,20]. However, to the authors knowledge the impact of the current level on the load angle is not described yet in literature and is therefore a novel contribution in this paper.

The load is modeled using lumped parameters: inertia J, viscous damping b and a load torque T_{load}:

$$T_{\text{motor}} - T_{\text{load}} = J\ddot{\theta} + b\dot{\theta} \tag{7}$$

In research on stepper motor motion control [31–33], it is common to simplify (1) by modelling the torque, load angle relation as:

$$T_{\text{motor}} = C_T.i_s.\sin(\delta) \tag{8}$$

It is important to note that this is not simply neglecting the second term in (1), meaning C_T is not equal to Ψ_r. Instead, C_T can be determined based on the measurements depicted in Fig. 4 using least squares to minimise the modelling error. Substituting (8) in (7) gives:

$$C_T.i_s.\sin(\delta) - T_{\text{load}} = J\ddot{\theta} + b\dot{\theta} \tag{9}$$

As shown in Fig. 3 the rotor postion θ can be written as the difference between the angular postion of the stator current vector β and the load angle δ:

$$C_T.i_s.\sin(\delta) - T_{\text{load}} = J(\ddot{\beta} - \ddot{\delta}) + b(\dot{\beta} - \dot{\delta}) \tag{10}$$

The current level i_s or magnitude of the stator current vector is used by the controller to obtain the load angle setpoint. Therefore the dynamic relation between the current level i_s and load angle δ is of interest. Eq. (10) shows the highly nonlinear character. However, the electromagnetic torque equation can be linearised around operating point (i_s^*, δ^*)

$$T_{\text{motor}} = C_T.i_s.\sin(\delta) \tag{11}$$

$$T_{\text{motor}} = T^*_{\text{motor}} + \left.\frac{\partial T_{\text{motor}}}{\partial i_s}\right|_* (i_s - i_s^*) + \left.\frac{\partial T_{\text{motor}}}{\partial \delta}\right|_* (\delta - \delta^*) \tag{12}$$

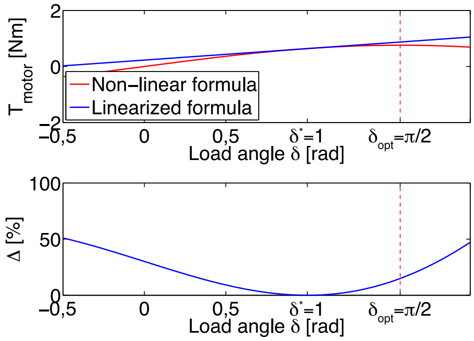


Fig. 10. Sensitivity analysis on the linearisation of the torque formula (11) .

$$T_{\text{motor}} = T^*_{\text{motor}} + C_T.\sin(\delta^*).\bar{i_s} + C_T.i^*_s.\cos(\delta^*).\bar{\delta} \tag{13}$$

A sensitivity analysis is used to assess the impact of this linearisation. Therefore, the deviation between the linearised (13) and nonlinear (11) torque equation is plotted in Fig. 10 for a common δ^* value of 1 rad. The deviation is expressed as a fraction of the nominal torque:

$$\Delta = \frac{T^{\text{linear}}_{\text{motor}} - T^{\text{nonlinear}}_{\text{motor}}}{T_{\text{nom}}}$$

The deviation depicted in Fig. 10 has a maximum of 50%. However, as discussed further in Section 7 and Fig. 20 the controller is disabled if the load angle increases too far above the setpoint δ^*. This certainly is the case if the load angle exceeds the optimal level $\delta_{\text{opt}} = \pi/2$rad. Furthermore, according to simulations (Fig 16) the load angle at maximum current for low speed and high load is already 0.25 rad. Between these boundaries of $\delta^* \in [0.25; \pi/2]$ the deviation between the linearized (13) and nonlinear (11) torque equation is maximum 19%. However, when the load angle approaches the setpoint δ^* this deviation drops rapidly. For instance, when the load angle δ only deviates 10% from the setpoint δ^* the deviation is limited to maximum 0.42%.

Eq. (13) is used instead of (11) to simplify Eq. (10):

$$T^*_{\text{motor}} + C_T.\sin(\delta^*).\bar{i_s} + C_T.i^*.\cos(\delta^*_s).\bar{\delta} - T_{\text{load}} = J\ddot{\beta} - J\ddot{\delta} + b\dot{\beta} - b\dot{\delta} \tag{14}$$

If steady-state is assumed, the electromagnetic motor torque T^*_{motor} equals the sum of the load T_{load} and friction $b\dot{\beta}$ torque. Furthermore, the acceleration $\ddot{\beta}$ of the current excitation vector $\mathbf{i}_s$ is zero as the estimation and control is done at constant imposed speed setpoints. In order to obtain a model which is valid for transient situations where the speed setpoint is constant but the real speed varies, the load angle acceleration $\ddot{\delta}$ is taken into account in the linearised model.

$$C_T.\sin(\delta^*).\bar{i_s} + C_T.i^*_s.\cos(\delta^*).\bar{\delta} = -J\ddot{\delta} - b\dot{\delta} \tag{15}$$

In the Laplace s-domain this yields:

$$\delta(s) = \frac{-C_T.\sin(\delta^*).I_s(s)}{C_T.i^*_s.\cos(\delta^*) + bs + Js^2} \tag{16}$$

The steady-state current i^*_s is determined by considering (9) in regime:

$$C_T.i^*_s.\sin(\delta^*) = b.\omega^* + T^*_{\text{load}} \tag{17}$$

This means (16) can be written as:

$$\frac{\delta(s)}{I(s)} = \frac{-C_T.\sin(\delta^*)}{\frac{b\omega^* + T^*_{\text{load}}}{\tan(\delta^*)} + bs + Js^2} \tag{18}$$

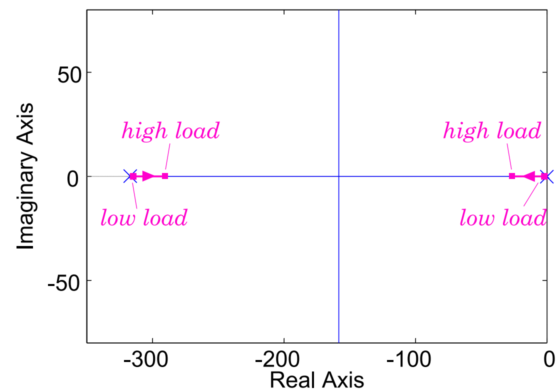


Fig. 11. Root locus describing the load angle dynamics as a function of the load $b.\omega^* + T^*_{\text{load}}$.

The process dynamics are characterised by the location of its poles being the roots of the denominator or characteristic equation in (18).

4.1. Load impact

As indicated by the denominator of (18), the process dynamics depend on i.a. the sum of the friction torque $b.\omega^*$ and the load torque T^*_{load} which can be seen as the total load. With the motor and test bench data given in Fig. 7 the characteristic equation is written as:

$$1 + \left(0,02.\omega^* + T^*_{\text{load}}\right).\frac{1}{1,61.10^{-4}s^2 + 0,051s} = 0 \tag{19}$$

The root locus of Fig. 11 shows the system dynamics ranging from an unloaded machine at low speed ($n^* = 50$tr/min) up to full load ($n^* = 500$tr/min, $T^*_{\text{load}} = 0,49$Nm). Fig. 11 shows that the load angle responds faster on current level changes for a more loaded machine at higher speeds. In this case, the load angle response is 12.5 times faster for the fully loaded machine ($n^* = 500$tr/min, $T^*_{\text{load}} = 0,49$Nm) compared to the unloaded machine at low speed ($n^* = 50$tr/min).

4.2. Estimator impact

The dynamic behaviour of the load angle estimator can be compared to the process dynamics. As mentioned before, the SDFT approach updates the estimated load angle value every measurement sample. However, the estimation is only completely updated after one full signal period. This dynamic estimator behaviour can be modelled as a first order system with a settling time equal to one full signal period:

$$\frac{1}{\tau s + 1} \tag{20}$$

The settling time of a first order system is 4 times the time constant τ, meaning:

$$T_{signal} = \frac{2\pi}{\omega p} = 4.\tau \tag{21}$$

In Fig. 12 both the system and estimator poles are depicted. The estimator impact is largest when the speed is low. However, even at 35 rpm, or 5% of the maximum speed, the estimator pole will be 4 times faster compared to the slowest system pole at high load. When the speed increases to 100rpm the estimator pole is already more than 11 times faster then the slowest system pole at high load.

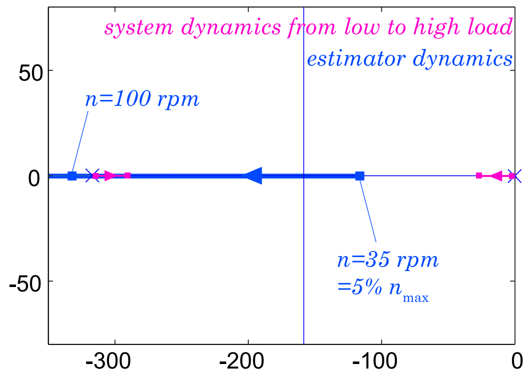


Fig. 12. Estimator and system dynamics represented by the location of their poles.

5. Load angle controller

Applying vector control schemes typical for position control of PMSM and described in [5,8,9] would result in a load angle control for stepping motors as depicted in Fig. 13. In such scheme's a cascade controller with a P- position controller and PI- speed controller is usually used to determine the current setpoint i_s^* while the load angle is controlled by adapting the angular position β of the stator current vector $\mathbf{i}_s$. In that case the load angle dynamics are only determined by the, usually fast, current controller dynamics and the load angle controller dynamics. This dynamic behaviour can easily be made faster than the dynamical relationship between stator current level i_s and load angle δ as described in the previous Section 4. However, the control architecture in Fig. 13 implies that at least 5 control parameters (P position control, P & I speed control and P & I load angle controller) should be designed. Furthermore, this control architecture does not only need the estimated load-angle $\hat{\delta}$ but also the position θ and speed n. This asks for an encoder or more demanding estimator. On the other hand, in industry stepping motor position is usually controlled in open-loop as illustrated in Fig. 5. For this open-loop control, step command pulses sent by the user determine the angular position β of the stator current vector $\mathbf{i}_s$. Therefore, the control structure depicted in Fig. 14 is proposed. This architecture means only two control parameters (P & I load angle controller) should be designed, the proposed load-angle estimator can be used and the algorithm can be implemented in currently running industrial stepping motor applications.

Based on the nonlinear relationship between the current amplitude i_s and the load angle δ, given in (1), one could conclude that advanced algorithms are necessary to control the load angle [34]. However, Section 4 shows that the linearized process dynamics are represented by two (real) poles. This means a straightforward PI controller is worth considering. It makes sense to start the stepper motor operation for a certain speed setpoint at maximum current. As soon as the load angle estimation becomes available, the current level can be reduced until the load angle has reached its setpoint δ^*. Fig. 15 shows such a current reduction. This is obtained using a PI controller for which the initial state of the Integrator equals the nominal current level. Also, both K_i and K_p have to be negative as the current level has to be reduced if the load angle needs to increase. This proposed sensorless control structure is depicted in Fig. 14.

The PI controller can be written in the s-domain:

$$K_p\left(\frac{s+1/T_i}{s}\right) \tag{22}$$

The denominator of (18) can be factorised into two real poles $(-a_1, -a_2)$ with $(a_1 \gg a_2)$. Meaning the process dynamics can be written as:

$$\frac{\delta(s)}{I(s)} = \frac{-C_T.\sin(\delta^*)}{J(s+a_1)(s+a_2)} \tag{23}$$

The controller zero can be used to compensate the slowest process pole $(s+a_1)$ [35]:

$$T_i = \frac{2.\dfrac{b\omega^* + T^*_{\text{load}}}{\tan(\delta^*)}}{-b+\sqrt{b^2 - 4J\dfrac{b\omega^* + T^*_{\text{load}}}{\tan(\delta^*)}}} \tag{24}$$

The resulting closed-loop dynamics can be written as:

$$\frac{\dfrac{-K_pC_T}{J}\sin(\delta^*)}{s^2 + a_2.s + \dfrac{-K_pC_T}{J}\sin(\delta^*)} \tag{25}$$

Given the standard second order process dynamics:

$$\frac{T}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{26}$$

K_p can be determined by comparing (26) and (25):

$$K_p = -\frac{J.\left(\dfrac{-b-\sqrt{b^2 - 4J\dfrac{b\omega^* + T^*_{\text{load}}}{\tan(\delta^*)}}}{2.\dfrac{b\omega^* + T^*_{\text{load}}}{\tan(\delta^*)}}\right)^2}{4.\zeta^2.C_T.\sin(\delta^*)} \tag{27}$$

Where ζ is the damping coefficient. Both K_p (27) and T_i (24) depend on the sum of the friction and load torque: $b\omega^* + T^*_{\text{load}}$. This total mechanical load is in most cases unknown and difficult to determine. Moreover, the load can vary over time. However when the load is driven at a constant speed setpoint and at maximum current $I_{\max}$ (9) becomes:

$$b\omega^* + T^*_{\text{load}} = C_T.I_{\max}.\sin\left(\delta_{i_{\max}}\right) \tag{28}$$

As mentioned before the stepper motor operation is started at maximum current for every new speed setpoint. This means the friction and load torque can be determined based on the estimated load angle for a stepper motor driven at maximum current $\delta_{i_{\max}}$. Finally K_p and T_i can be written as:

$$K_p = -\frac{J.\left(\dfrac{-b-\sqrt{b^2 - 4JC_T.I_{\max}.\dfrac{\sin(\delta_{i_{\max}})}{\tan(\delta^*)}}}{2.C_T.I_{\max}.\dfrac{\sin(\delta_{i_{\max}})}{\tan(\delta^*)}}\right)^2}{4.\zeta^2.C_T.\sin(\delta^*)} \tag{29}$$

$$T_i = \frac{2.C_T.I_{\max}.\dfrac{\sin(\delta_{i_{\max}})}{\tan(\delta^*)}}{-b+\sqrt{b^2 - 4JC_T.I_{\max}.\dfrac{\sin(\delta_{i_{\max}})}{\tan(\delta^*)}}} \tag{30}$$

6. Simulations

According to the design procedure, described in the previous section, the damping ζ is chosen by the user to obtain the desired load angle behaviour. Whenever the actual load angle δ exceeds the stable operating range $\left(\delta \in [-\delta_{\text{opt}}, +\delta_{\text{opt}}]\right)$ due to oscillations or overshoot, step loss leads to loss of synchronism. Therefore the damping ratio ζ needs to be sufficiently high to guarantee stable operation as illustrated by simulation results in Fig. 16 a and b. On the other hand, an overly damped controller, as depicted in

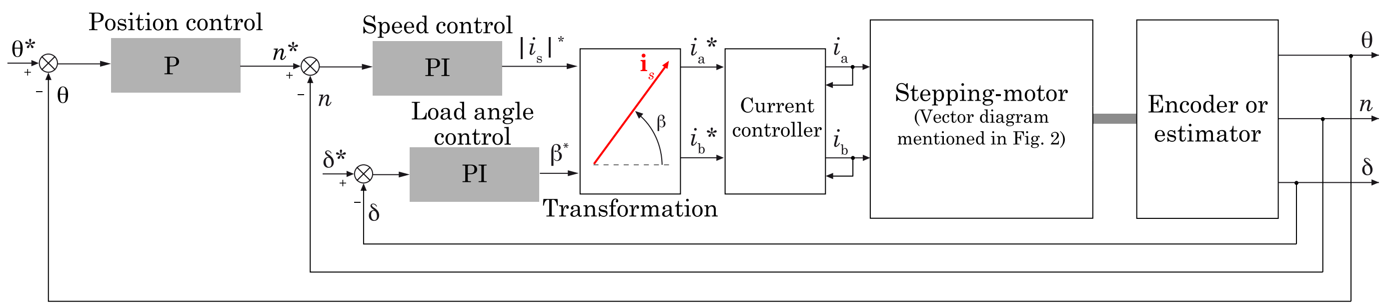


Fig. 13. Typical PMSM vector control structure applied to a two-phase hybrid stepping motor.

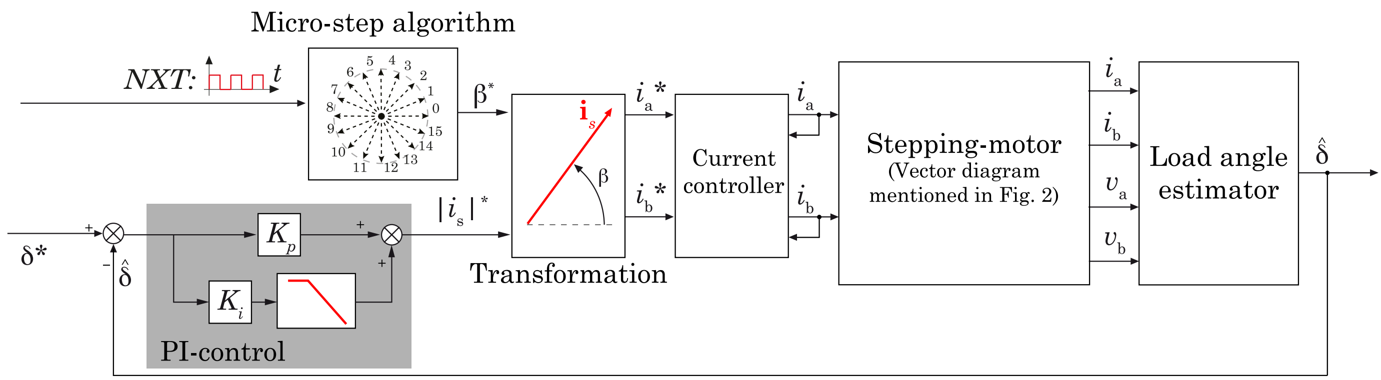


Fig. 14. Proposed sensorless control structure.

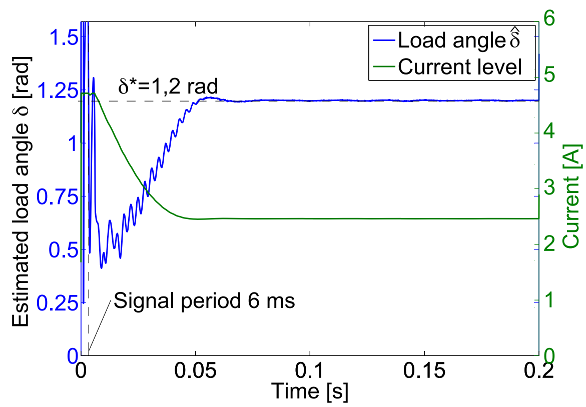


Fig. 15. PI load angle control, $K_i = -150$ and $K_p = -0.05$ for 200 rpm and load torque 0.124Nm.

Fig. 16 c, leads to a slow system. A higher load angle setpoint δ^* is closer to the stability boundary δ_{opt} meaning a higher damping will be necessary to obtain stable control. This effect is illustrated by the simulation results in Fig. 16 c and d.

The fact that a sufficiently high damping coefficient ζ is needed to avoid overshoot and subsequently step loss is illustrated by the root locus depicted in Fig. 17a. This root locus is obtained by using the proposed controller design procedure. This figure shows the closed-loop poles move further away from the critical point, where overshoot occurs, for higher damping ratio ζ values.

For industrial implementation robustness of the controller is essential. Especially robustness for mechanical load variations are of interest. As indicated by Eq. (28) the friction and load torque can be determined based on the estimated load angle at maximum current $\delta_{i_{max}}$. However, the load inertia also affects the controller settings defined in Eqs. (29) and (30). Therefore the impact of an erroneous estimated load inertia on the controller performance should be assessed.

In literature it is common to consider a load inertia variation between 1/10th of the motor inertia and 10 times the motor inertia [36]. While the total inertia used for the simulation is $J = J_{motor} + J_{load} = 2.2\,10^{-5}\text{kgm}^2 + 2.2\,10^{-5}\text{kgm}^2$ the controller design is based on an erroneous inertias of $J = J_{motor} + 0.1J_{load} = 2.42\,10^{-5}\text{kgm}^2$ (Root locus in Fig. 17b), $J = J_{motor} + 10J_{load} = 24.2\,10^{-5}\text{kgm}^2$ (Root locus in Fig. 17c) and the correct inertia $4.4\,10^{-5}\text{kgm}^2$ (Root locus in Fig. 17a). The closed-loop performance of the load angle controller designed based on erroneous inertias is depicted in Table 2. Next to typical time domain performance such as settling time T_s and procentual overshoot, the damping ζ_{crit} at which overshoot and consequently step loss occurs is given in Table 2. Using the described distorted inertia for controller design does not result in overshoot or instabilities as shown in Table 2.

Comparing the critical damping ratio ζ_{crit} with the damping ratio $\zeta = 2$ used for the controller design gives an idea of the stability margin of the controller. This margin is also given in Table 2 as the ratio of the K_p used in the controller and the K_p value leading to overshoot, step loss and instability. If the controller is based on an inertia J_{design} lower than the correct one J_{mod} this results in a faster system with a lower margin for ζ and K_p. When on the contrary the inertia used for the controller design J_{design} is higher than the correct one J_{mod} the system becomes slower while the margin for ζ and K_p increases. In both cases the load angle controlled system is still stable.

7. Measurements

Implementing the proposed sensorless load-angle control involves the following major steps:

- The voltage v and current i signals between the drive and the motor should be measured. As classical stepping motor drive methods involve current control the current signals should already be measured by the drive. This means there is no need for new current sensors. The voltage on the other hand is typically not measured by the drive. However, the voltage signal

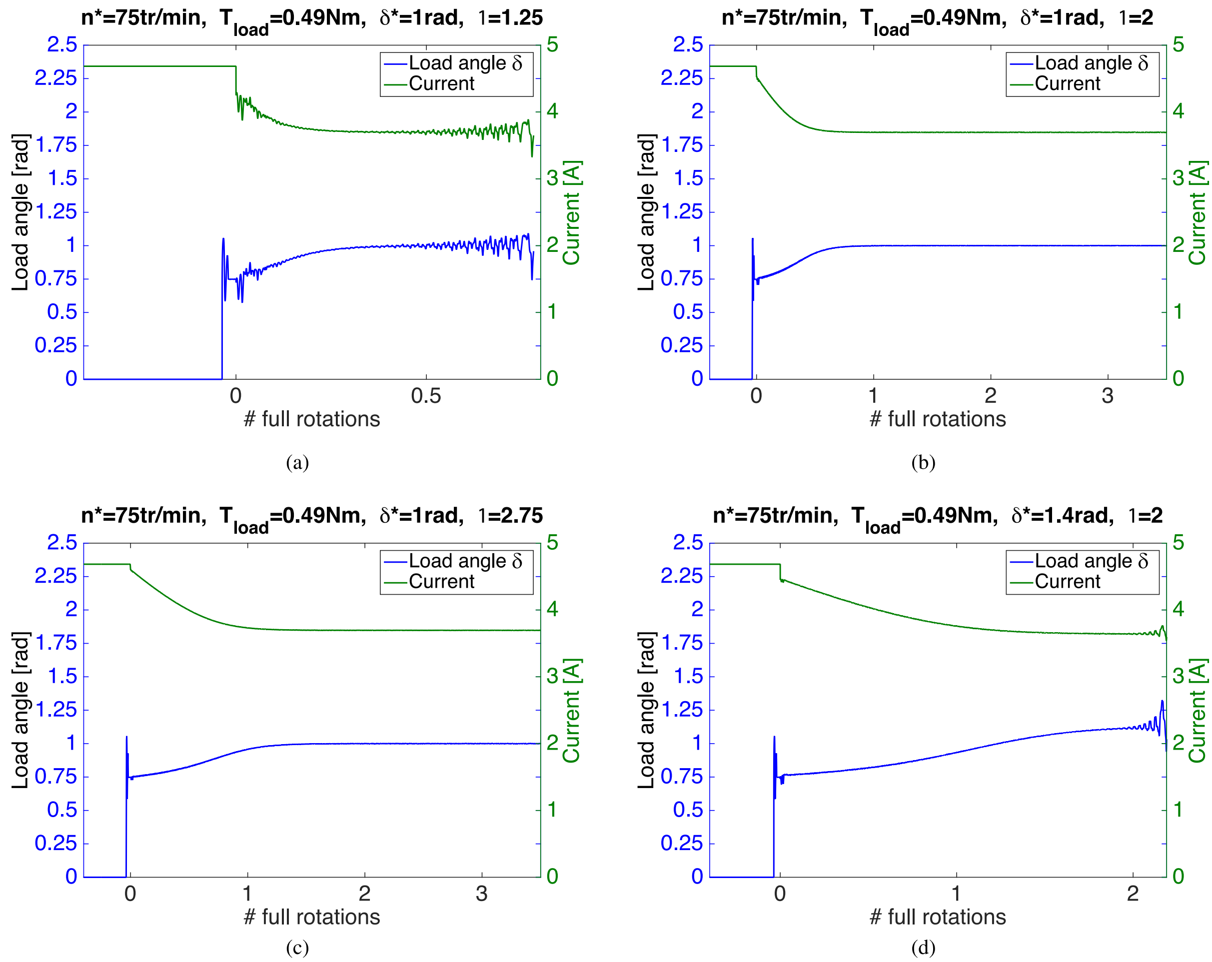


Fig. 16. Simulation of the controlled current reduction and estimated load angle at $n^* = 75$rpm and $T^*_{\text{load}} = 0.49$Nm for different load angle setpoints δ^* and damping coefficients ζ.

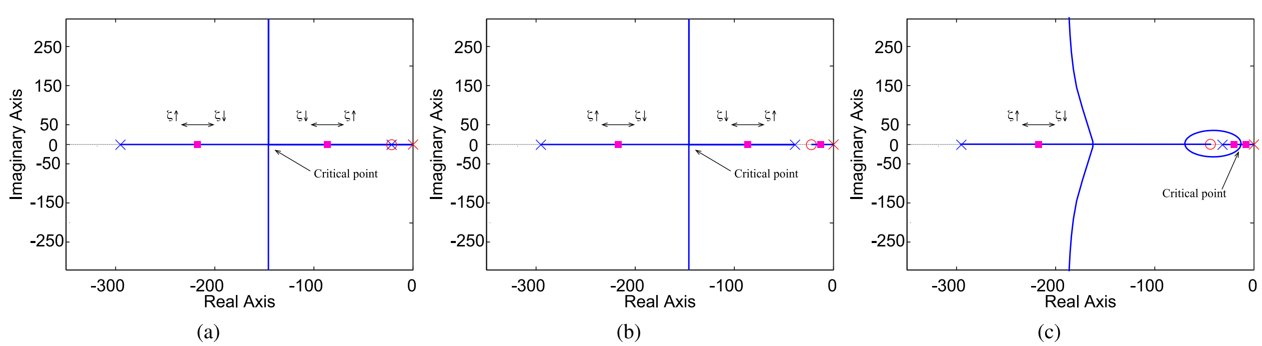


Fig. 17. Root locus including system and controller poles and zeros for $n^* = 75$rpm, $T^*_{\text{load}} = 0.49$Nm, $\delta^* = 1rad$, $\zeta = 2$.

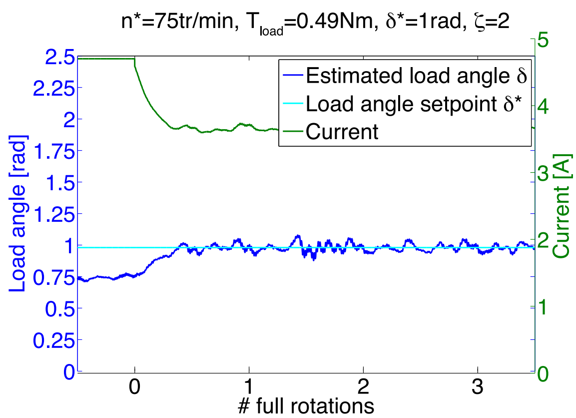
can be determined based on the states of the power electronic switching elements. In that case, the inverter non-linearities, distorting the estimated phase voltage, should be compensated as discussed in [37].

- The measured voltage and current signals should be processed by the SDFT algorithm as depicted in Fig. 6b. This transform can be implemented in the drive itself or a separate microcontroller. The results of the SDFT can be used in Eq. (6) to determine the estimated load angle $\hat{\delta}$.
- The estimated load angle can be used by the PI load angle controller as depicted in Fig. 14. While the micro-step algorithm, transformation and current controller blocks of Fig. 14 are typically included in the commercial stepping motor drives, the load angle estimator and PI controller can be implemented in a seperate microcontroller.

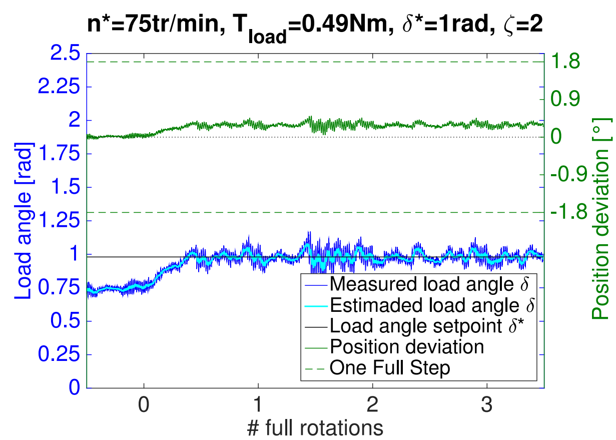
For this paper, the load angle control algorithm is implemented in the dSpace controller used in the test rig of Fig. 7. Fig. 18a shows the controlled current reduction to obtain the load angle setpoint $\delta^* = 1$rad at low speed $n^* = 75\text{rpm} = 15\% n_{\max}$ and maximum load $T_{\text{load}} = 0.49$Nm. Using a damping $\zeta = 2$ the current is reduced to 77% of its original level in 78 full-step commands (0.39 of a full rotation). Fig. 19a on the other hand shows the controlled current reduction to obtain the load angle setpoint $\delta^* = 0,6$rad at maximum speed $n^* = 500$rpm and no load. In this case, using a damp-

Table 2
Robustness against load inertia variations between the modelled ineratia J_{mod} and values used for the controller design J_{design}.

	$J_{design} = J_{motor} + 0.1J_{load}$	$J_{design} = J_{mod} = J_{motor} + J_{load}$	$J_{design} = J_{motor} + 10J_{load}$
Closed-loop zero locations	-5.1	-5.1	-5.4
Closed-loop pole locations	$-388, -59.2, -5.1$	$-417, -30, -5.1$	$-443, -4.8 \pm 1.1i$
ζ_{crit}	1.4	1.4	0.4
T_s [s]	007	013	077
Overshoot [%]	0	0	0

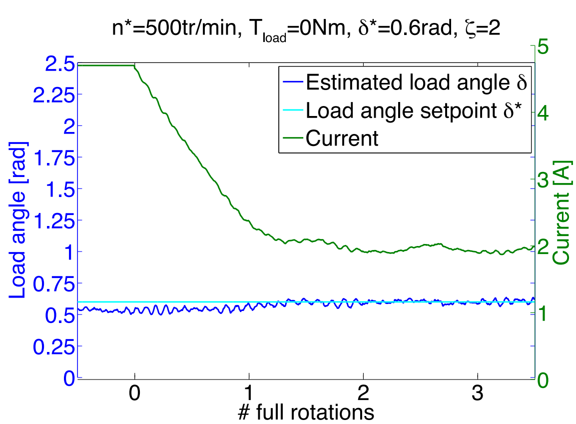


(a) Controlled estimated load angle and drive current

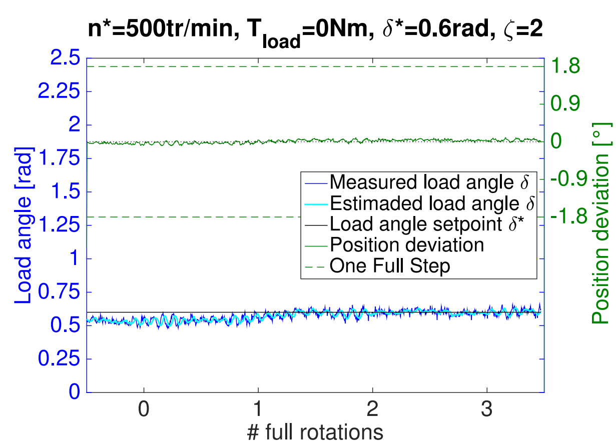


(b) Position deviation, estimated and measured load angle

Fig. 18. Measurement of the controlled current reduction and load angle optimisation at low speed (15% n_{max}) and full load.

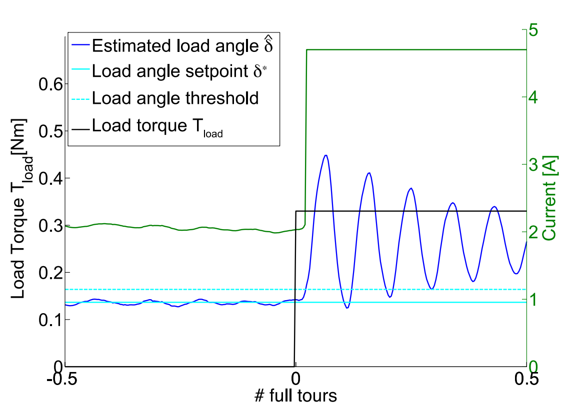


(a) Controlled estimated load angle and drive current

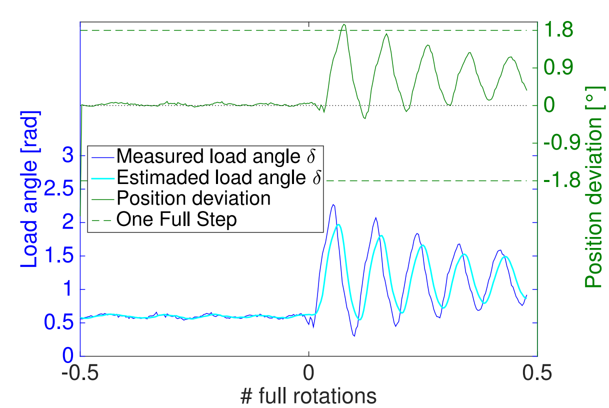


(b) Position deviation, estimated and measured load angle

Fig. 19. Measurement of the controlled current reduction and load angle optimisation at maximum speed and no load.



(a) Controlled estimated load angle and drive current



(b) Position deviation, estimated and measured load angle

Fig. 20. Load angle threshold is exceeded when the load torque suddenly increases from no load to 0.33Nm, resulting in a reset of the drive current i_s to its maximum value.

Table 3
Current reduction and efficiency gain, when using sensorless load angle control at different operating points.

n^* [rpm] (n_{max}=500rpm) T_{load} [Nm] (T_{max}=0.49Nm)	75† 0.49†	210 0.3	350 0.3	490 0.45	500† 0†
i_{red} [compared to i_{max}]	77%	47%	67%	83%	42%
η at i_{max}	20%	24%	32%	58%	⁄
η at i_{red}	30%	54%	45%	59%	⁄

†Depicted in Figs. 18 and 19.⁄ There is no output power for a no loaded machine.

Table 4
Maximum load torque increase [38].

	Regime situations				
Rotational speed [rpm] Torque [Nm]	75 0	75 0.49	375 0.33	500 0	500 0.49
Max ΔT_{load} if δ control is used together with (31)	0.17	0.07	0.23	0.33	0.04
Max ΔT_{load} for continous operation at i_{max}	0.5	0.13	0.23	0.36	0.06

ing $\zeta = 2$ leads to a current reduction up to 42% of the nominal current after 260 full-steps (1,3 full tours).

Both Figs. 18 and 19 also show the actual measured load angle value and positie deviation on the left. As illustrated in Figs. 18b and 19 b the load angle estimation is less noisy compared to the real measured load angle thanks to the SDFT which only considers the fundamental signal components. From the users viewpoint the angular displacement after each step command pulse is important. This angular displacement, determined by the micro stepping setting, is constant for constant load angle values. However, when the load angle alters during a positioning task, small positioning errors could arise at the end of the trajectory. These errors are depicted in Figs. 18b and 19 b as well. Fig. 18b shows a final error of 0.29° for the current reduction of 77%. In Fig. 19b shows a very minimal positioning error for the current reduction of 42%.

Table 3 summarises the possible current reduction for the measurements depicted in Figs. 18 and 19 and three additional operating points. In addition the energy efficiency η measured when the motor is driven at the maximum current i_{max} and driven at reduced current i_{red} is added. Whereas other papers focus on torque ripples, the results in Table 3 show the energy-efficiency can be drastically increased by using the sensorless load angle controller to reduce the drive current. Most controllers described in literature adapt both the current amplitude and angular position. This means positioning by step command pulses sent by the user is no longer possible. However, in this paper only the current amplitude is changed meaning the user can still use step command pulses for position control. Moreover, the estimator does not need any tuning and operates load independent.

A detailed dynamical analysis discussed in 4 proved theoretically the estimator dynamics are sufficiently fast to track rapidly varying load torques. To achieve a robust control the drive current i_s could be reset at its maximum value i_{max} whenever a load angle threshold value δ_{safe} is exceeded:

$$i_s = i_{max} : \begin{cases} \text{SET :} & \hat{\delta} \geq \delta_{safe} \\ \text{RESET :} & \hat{\delta} \leq \delta^* \end{cases} \tag{31}$$

The optimal value for δ_{safe} at which the PI load angle controller is bypassed as defined by (31) is determined in [38]. Based on [38] a load angle δ_{safe} threshold of 120% of the load angle setpoint δ^* is used in this paper. Fig. 20 depicts the measurement result at such a load torque disturbance. The bypass mechanism of (31) clearly avoids loss of synchronism between the actual rotor position and the step command pulses after a sudden load torque increase. Due to the sudden load angle increase and subsequent controller reaction the positioning error (Fig. 20b) becomes large but this effect damps when the positioning task is continuing.

The loss of robustness due to load angle optimisation and subsequent current reductions as summarized in Table 3 can be counteracted by equation (31). However, Table 4 from [38] shows the loss of robustness by applying a load angle controller with (31) as a safety measure. Table 4 shows both the maximum load torque increase for a load angle controlled situation and a situation where the current level is always kept at its maximum $i_s = i_{max}$.

As mentioned before, the current will also be reset to its maximum value when a new speed setpoint is applied.

8. Conclusions

The load angle determines the torque/current ratio of the stepper motor drive. In this paper, a previously described load angle estimator is used. This estimator is based on fundamental components of the measured phase current and voltage. Moreover, the estimator does not need tuning and only relies on easily measurable electrical parameters. This implies the load has no influence on the estimator. Using the SDFT the estimator can be implemented with a low computational cost. Furthermore, as the estimations are based on the regular electrical signals, no signal injection is needed. Hence, the estimator does not need to have control over the power electronics meaning the estimator is implementable in current stepping motor driven systems.

In this paper, the estimated load angle is used in a feedback loop to obtain an optimal torque/current ratio. In order to obtain a robust load angle control, the system dynamics are profoundly discussed. In this way the current can be reduced from the nominal level to the minimum current necessary to drive the motor at a specific load torque and speed setpoint. Both simulation results and measurements on the practically implemented algorithm are included in this paper for validation. Robustness of the controller is guaranteed as load parameters essential for the controller design can be derived based on the estimated load angle. The impact of the inertia, which is harder to identify based on the estimated load angle, is analysed in this paper to prove its limited impact on the stability of the load angle controller.

Determination of the commutation or angular position of the stator current vector by means of step command pulses is very common in stepper motor applications. This research proposes a closed-loop controller which is complementary to this approach. In contrast to more complex controllers, the algorithm proposed in this paper only adjusts the current level to obtain the desired load angle. In other words, the controller determines the amplitude of the stator current vector while the position of this vector is determined by step commands. Together with the fact that a PI-control structure is used this guarantees the industrial employability of the algorithm.

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